Data Analyst Job Market Analysis Report

Salary Trends, Experience Requirements, and Programming Language Demand

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Dataset: Data Analyst Job Market Analysis — Sample Postings (n = 400)

Analyst: Nola Vanualailai

1. Introduction

This report analyses 400 data analyst job postings to explore the relationships between **median salary estimates**, **minimum years of experience required**, and **demand for programming languages** (specifically R and/or Python). The dataset was extracted from online job postings using Large Language Models (LLMs) within Google Sheets as part of a generative AI workshop on automated data analysis.

The primary research questions guiding this analysis were:

- Does requiring more years of experience correlate with higher salary offers?
- How does demand for Python versus R (or both) relate to salary levels?
- What does the distribution of salary and experience requirements reveal about the current data analyst job market?

Key Findings at a Glance: The overall median salary is **$68,000**, with a wide range from $33,500 to $150,000. Minimum years of experience shows almost no linear relationship with salary (correlation = 0.051). Python is the most in-demand programming language among postings that specify a requirement, while roles requiring both Python and R command the highest average salaries. Notably, 63.8% of postings do not explicitly require either language, yet their salary distribution remains comparable.

2. Years of Experience Analysis

Methodology

A scatter plot was generated using the 310 postings that contained valid numeric experience requirements (90 postings with missing experience data were excluded from this specific analysis). A linear trend line was fitted to assess the direction and strength of the relationship between minimum years of experience and median salary estimate.

Key Findings

- **Correlation (Pearson r):** 0.051 — indicating a negligible positive linear relationship.
- **Trend slope:** Approximately +$573 per additional year of required experience.
- High-salary outliers (above $120,000) appear at nearly every experience level from 0 to 8 years.
- The densest cluster of postings falls between **2–5 years** of experience and **$50,000–$85,000** in salary.

Interpretation

The scatter plot reveals a very weak positive relationship between the minimum years of experience required and median salary. While the trend line slopes slightly upward, the wide spread of data points shows that experience alone is a poor predictor of salary. Many postings requiring 0–3 years of experience offer salaries above $100,000, while some roles asking for 8–10 years offer comparatively modest compensation. This suggests that other factors — such as industry, company size, location, specific technical skills, or negotiation — play a much stronger role in determining salary than years of experience. A key limitation is that 90 postings (22.5%) lacked experience data and were excluded; the small sample size for very senior roles (n < 20 for 10+ years) also means trends at the upper end should be interpreted cautiously.

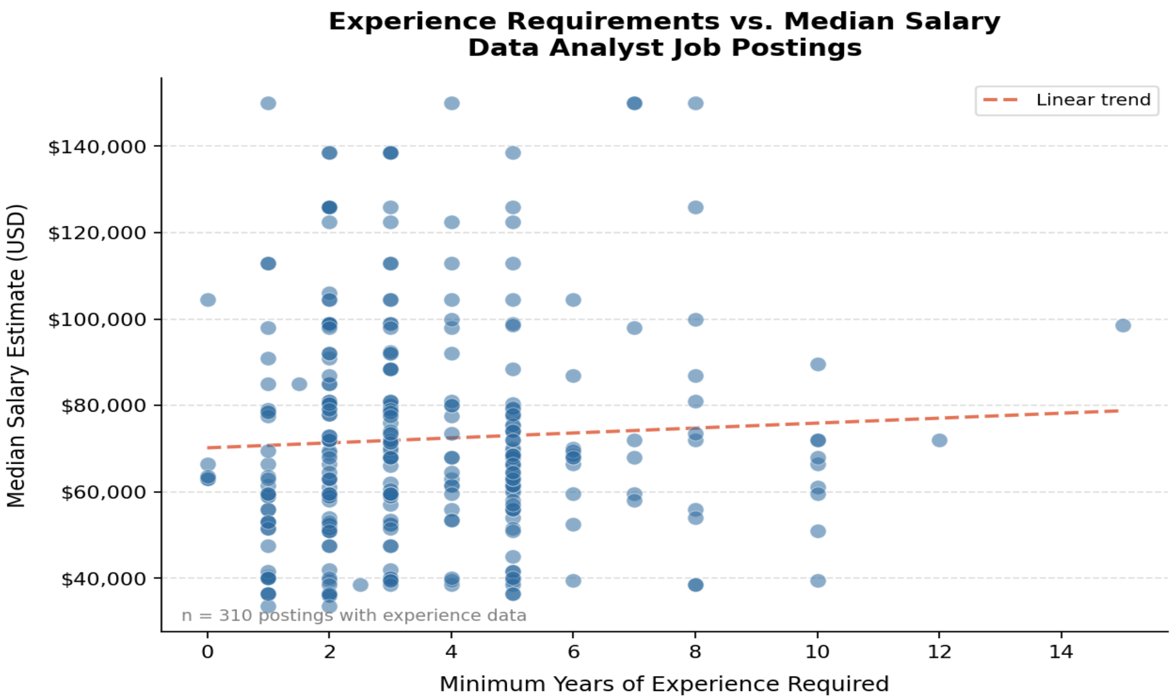

Figure 2.1: Salary vs. Minimum Years of Experience Required (n = 310)

3. Programming Language Analysis

Methodology

A box plot was created using all 400 postings to compare salary distributions across four standardised programming language categories: **R only**, **Python only**, **Both (R and Python)**, and **Neither**. Box plots allow clear comparison of medians, interquartile ranges, and outliers across groups.

Key Findings

- **Both (R and Python):** Highest mean salary (~$75,900) and median of $69,000, with the widest spread.
- **Python only:** Median salary of $69,000 — nearly identical to the 'Both' category.
- **Neither:** Median salary of $68,000 — surprisingly comparable to language-specific roles.
- **R only:** Lowest median salary ($61,500) and least variability, though based on only 15 postings.
- High-salary outliers (above $120,000) exist across all four categories.

Interpretation

The box plot shows only modest differences in salary distributions across language categories. While roles requiring both R and Python have a slightly higher mean, the overlap between all groups is substantial. The difference between the highest median ($69,000 for Both/Python) and lowest ($61,500 for R only) is just $7,500. This indicates that programming language requirements alone are a relatively weak predictor of salary in the current data analyst market. Other factors such as industry, seniority, and specific domain expertise likely matter more. *Limitation:* The R-only category contains only 15 postings; its results should be treated as indicative rather than generalisable.

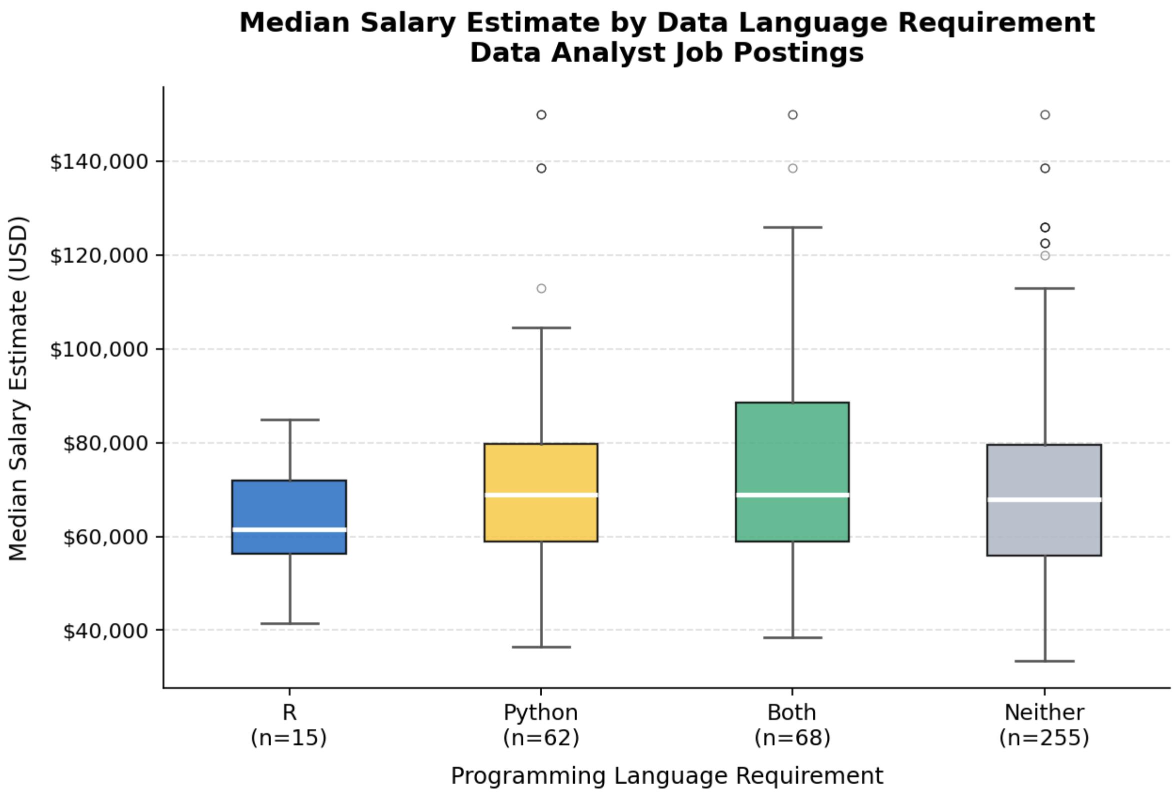

Figure 3.1: Median Salary Distribution by Programming Language Requirement

4. AI Reflection

I need to be honest about what I thought before I started this project. In Fiji, where I have worked in public health, the pay system is quite straightforward. You put in the years and you move up the scale. A nurse with five years gets more than one with two. A doctor with ten years is on a different band entirely. So when I first looked at this salary data, I assumed the same pattern would appear — more experience equals more money. It turns out that assumption does not really hold here, and I am still processing what that means.

Analysis 1 – Experience vs Salary

The correlation of 0.051 surprised me. I ran it twice because I thought I had made a mistake. In the health surveys I work on back home, a number that low would mean there is basically no relationship — and that is exactly what the scatter plot showed. What really stood out was seeing salaries over $120,000 for jobs that wanted zero years of experience. In Fiji, that would almost never happen unless you were a specialist returning from overseas with years of advanced training. Here it seemed almost normal, and that challenged my assumptions.

I think the difference comes down to structure. Fiji's health system operates under clear public-sector pay bands. Everything is predictable and transparent. This dataset felt messy by comparison.

Employers seem to list “minimum years” as a rough guideline rather than a strict rule tied to salary. I kept asking myself: is this flexibility, or is it just chaos? I still don’t have a clear answer.

I was also bothered by the 90 postings that had no experience data at all. In Fiji, every Ministry of Health job advertisement clearly lists requirements. Seeing nearly a quarter of this dataset missing that information made me wonder about the employers behind those postings. I kept thinking that if I had those 90 records, my conclusions might shift. I will never know, and that uncertainty stayed with me.

Analysis 2 – Programming Languages

This part felt personal. I have been learning R for health data analysis and had only recently started learning Python. I was quietly hoping the analysis would show that the time I invested in Python was financially worthwhile. When I saw that the “Neither” category had almost the exact same median salary as Python-only roles, it gave me pause. I sat there thinking about the nights spent learning R, fixing do-files, and merging national health survey datasets. Was that effort mainly about earning more money, or was it about being able to do my job independently without waiting for overseas consultants? I think it was the second reason — and that is perfectly valid, even if it wasn’t what I was measuring here.

What I am taking away

The biggest lesson for me is that I need to stop treating the Fijian health system as a universal template for how labour markets work. That system is structured, predictable, and transparent. The market represented in this dataset is none of those things. Experience does not reward people in the way I expected. Programming languages do not separate candidates as cleanly as I thought. Everything is more interconnected and less linear than I assumed.

If I do more analysis like this in future, I want to split the data by actual job title and industry first. “Data analyst” and “data scientist” are not interchangeable, just as “nurse” and “doctor” are not the same. I treated them as similar and that was an oversimplification.

This project was genuinely good for me. It was uncomfortable at times because the data pushed back against my assumptions. I am used to health data where patterns mostly behave as expected. This dataset did not. I am leaving with more questions than answers — and I think that is actually the point of doing real analysis rather than simply confirming what you already believe.